

## 4. Bud rate

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### 1 Environment preparation

Run scripts with the needed packages, functions, and the raw data frame.

```
source("0a_packages.R")
source("0b_script_data.R")
source("0c_functions.R")
```

### 2 Data formatting

Create a new data frame adapted for generalized linear mixed-effects models (GLMM).

```

buds_df <-

  #' Load the raw data frame.
  df %>%

  #' Reshape the weekly bud counts from wide to long format.
  pivot_longer(cols = starts_with("buds_"),
               names_to = c(".value", "week"),
               names_sep = "_",) %>%

  #' Convert week and bud count to integers.
  mutate(week = as.integer(week),
         buds = as.integer(buds),
  #' Create a centered variable for the week.
         week_centered = week - mean(week, na.rm = TRUE)) %>%

  #' Remove weeks with missing bud count.
  filter(!is.na(buds)) %>%

  #' Keep only the variables needed for the analysis.
  select(rest_condition, lineage, tum_state, week, week_centered,
         buds, id_unique, parent_unique, replicate_unique)

```

Set reference levels for the categorical variables.

```

buds_df$lineage <- relevel(factor(buds_df$lineage), ref = "HO_MT")
buds_df$rest_condition <- relevel(factor(buds_df$rest_condition), ref = "Control")
buds_df$tum_state <- relevel(factor(buds_df$tum_state), ref = "Healthy")

```

## 3 Model selection

### 3.1 Random effects selection

With the full fixed-effect structure held constant, compare different random-effect structures to find the optimal one.

```

buds_pois_R_00 <- glmmTMB(buds ~ week_centered * lineage * rest_condition, REML = TRUE,
family = poisson, data = buds_df)
buds_pois_R_01 <- glmmTMB(buds ~ week_centered * lineage * rest_condition + (1 |
id_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_02 <- glmmTMB(buds ~ week_centered * lineage * rest_condition +
(week_centered | id_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_03 <- glmmTMB(buds ~ week_centered * lineage * rest_condition + (1 |
parent_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_04 <- glmmTMB(buds ~ week_centered * lineage * rest_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_05 <- glmmTMB(buds ~ week_centered * lineage * rest_condition + (1 |
id_unique) + (1 | parent_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_06 <- glmmTMB(buds ~ week_centered * lineage * rest_condition + (1 |
id_unique) + (1 | replicate_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_07 <- glmmTMB(buds ~ week_centered * lineage * rest_condition + (1 |
id_unique) + (1 | parent_unique) + (1 | replicate_unique), REML = TRUE, family = poisson,
data = buds_df)

```

```
AIC(buds_pois_R_00, buds_pois_R_01, buds_pois_R_02, buds_pois_R_03, buds_pois_R_04,
    buds_pois_R_05, buds_pois_R_06, buds_pois_R_07) %>% arrange(AIC)
```

```
##           df      AIC
## buds_pois_R_04 25 12602.13
## buds_pois_R_06 26 12604.13
## buds_pois_R_07 27 12606.13
## buds_pois_R_02 27 12616.80
## buds_pois_R_03 25 12625.02
## buds_pois_R_01 25 12625.68
## buds_pois_R_05 26 12626.02
## buds_pois_R_00 24 12628.14
```

The model with the replicate id as a random effect (buds\_pois\_R\_04) has the lowest AIC, this random effect structure is therefore selected.

### 3.2 Fixed effects selection

With the selected random-effect structure fixed, compare different fixed-effect structures to find the optimal one.

```
buds_pois_F_00 <- glmmTMB(buds ~ 1 + (1 | replicate_unique), REML = FALSE, family =
poisson, data = buds_df)
buds_pois_F_01 <- glmmTMB(buds ~ week_centered + (1 | replicate_unique), REML = FALSE,
family = poisson, data = buds_df)
buds_pois_F_02 <- glmmTMB(buds ~ lineage + (1 | replicate_unique), REML = FALSE, family =
poisson, data = buds_df)
buds_pois_F_03 <- glmmTMB(buds ~ rest_condition + (1 | replicate_unique), REML = FALSE,
family = poisson, data = buds_df)
buds_pois_F_04 <- glmmTMB(buds ~ tum_state + (1 | replicate_unique), REML = FALSE, family
= poisson, data = buds_df)
buds_pois_F_05 <- glmmTMB(buds ~ week_centered + lineage + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_06 <- glmmTMB(buds ~ week_centered + rest_condition + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_07 <- glmmTMB(buds ~ week_centered + tum_state + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
buds_pois_F_08 <- glmmTMB(buds ~ lineage + rest_condition + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
buds_pois_F_09 <- glmmTMB(buds ~ lineage + tum_state + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_10 <- glmmTMB(buds ~ rest_condition + tum_state + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_11 <- glmmTMB(buds ~ week_centered * lineage + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_12 <- glmmTMB(buds ~ week_centered * rest_condition + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_13 <- glmmTMB(buds ~ week_centered * tum_state + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
buds_pois_F_14 <- glmmTMB(buds ~ lineage * rest_condition + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
```

```

buds_pois_F_15 <- glmmTMB(buds ~ lineage * tum_state + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_16 <- glmmTMB(buds ~ rest_condition * tum_state + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_17 <- glmmTMB(buds ~ week_centered + lineage + rest_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_18 <- glmmTMB(buds ~ week_centered + lineage + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_19 <- glmmTMB(buds ~ week_centered + rest_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_20 <- glmmTMB(buds ~ lineage + rest_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_21 <- glmmTMB(buds ~ week_centered * lineage + rest_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_22 <- glmmTMB(buds ~ week_centered * rest_condition + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_23 <- glmmTMB(buds ~ week_centered * tum_state + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_24 <- glmmTMB(buds ~ week_centered * lineage + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_25 <- glmmTMB(buds ~ week_centered * rest_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_26 <- glmmTMB(buds ~ week_centered * tum_state + rest_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_27 <- glmmTMB(buds ~ lineage * rest_condition + week_centered + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_28 <- glmmTMB(buds ~ lineage * tum_state + week_centered + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_29 <- glmmTMB(buds ~ rest_condition * tum_state + week_centered + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_30 <- glmmTMB(buds ~ lineage * rest_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_31 <- glmmTMB(buds ~ lineage * tum_state + rest_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_32 <- glmmTMB(buds ~ rest_condition * tum_state + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_33 <- glmmTMB(buds ~ week_centered * lineage * rest_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_34 <- glmmTMB(buds ~ week_centered * lineage * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_35 <- glmmTMB(buds ~ week_centered * rest_condition * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_36 <- glmmTMB(buds ~ lineage * rest_condition * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)

AIC(buds_pois_F_00, buds_pois_F_01, buds_pois_F_02, buds_pois_F_03, buds_pois_F_04,
buds_pois_F_05, buds_pois_F_06, buds_pois_F_07, buds_pois_F_08, buds_pois_F_09,
buds_pois_F_10, buds_pois_F_11, buds_pois_F_12, buds_pois_F_13, buds_pois_F_14,
buds_pois_F_15, buds_pois_F_16, buds_pois_F_17, buds_pois_F_18, buds_pois_F_19,
buds_pois_F_20, buds_pois_F_21, buds_pois_F_22, buds_pois_F_23, buds_pois_F_24,
buds_pois_F_25, buds_pois_F_26, buds_pois_F_27, buds_pois_F_28, buds_pois_F_29,
buds_pois_F_30, buds_pois_F_31, buds_pois_F_32, buds_pois_F_33, buds_pois_F_34,
buds_pois_F_35, buds_pois_F_36) %>% arrange(AIC)

```

| ## |                | df | AIC      |
|----|----------------|----|----------|
| ## | buds_pois_F_33 | 25 | 12489.87 |
| ## | buds_pois_F_21 | 14 | 12494.81 |
| ## | buds_pois_F_11 | 13 | 12500.31 |
| ## | buds_pois_F_24 | 14 | 12500.54 |
| ## | buds_pois_F_34 | 19 | 12509.36 |
| ## | buds_pois_F_27 | 14 | 12561.96 |
| ## | buds_pois_F_22 | 10 | 12563.02 |
| ## | buds_pois_F_17 | 9  | 12564.93 |
| ## | buds_pois_F_05 | 8  | 12570.68 |
| ## | buds_pois_F_18 | 9  | 12570.97 |
| ## | buds_pois_F_23 | 10 | 12572.75 |
| ## | buds_pois_F_28 | 11 | 12574.36 |
| ## | buds_pois_F_14 | 13 | 12862.38 |
| ## | buds_pois_F_30 | 14 | 12862.49 |
| ## | buds_pois_F_08 | 8  | 12866.84 |
| ## | buds_pois_F_32 | 10 | 12867.85 |
| ## | buds_pois_F_20 | 9  | 12868.47 |
| ## | buds_pois_F_31 | 11 | 12870.53 |
| ## | buds_pois_F_36 | 19 | 12871.30 |
| ## | buds_pois_F_02 | 7  | 12874.06 |
| ## | buds_pois_F_09 | 8  | 12874.72 |
| ## | buds_pois_F_15 | 10 | 12878.25 |
| ## | buds_pois_F_25 | 6  | 12881.20 |
| ## | buds_pois_F_29 | 6  | 12882.98 |
| ## | buds_pois_F_19 | 5  | 12883.40 |
| ## | buds_pois_F_35 | 9  | 12883.43 |
| ## | buds_pois_F_26 | 6  | 12885.35 |
| ## | buds_pois_F_07 | 4  | 12893.10 |
| ## | buds_pois_F_12 | 5  | 12894.87 |
| ## | buds_pois_F_13 | 5  | 12895.04 |
| ## | buds_pois_F_06 | 4  | 12896.94 |
| ## | buds_pois_F_01 | 3  | 12902.29 |
| ## | buds_pois_F_16 | 5  | 13173.44 |
| ## | buds_pois_F_10 | 4  | 13174.39 |
| ## | buds_pois_F_04 | 3  | 13186.02 |
| ## | buds_pois_F_03 | 3  | 13189.04 |
| ## | buds_pois_F_00 | 2  | 13195.82 |

The model with the interaction between the week, the lineage and the rest condition has the lowest AIC and is therefore selected (buds\_pois\_F\_33).

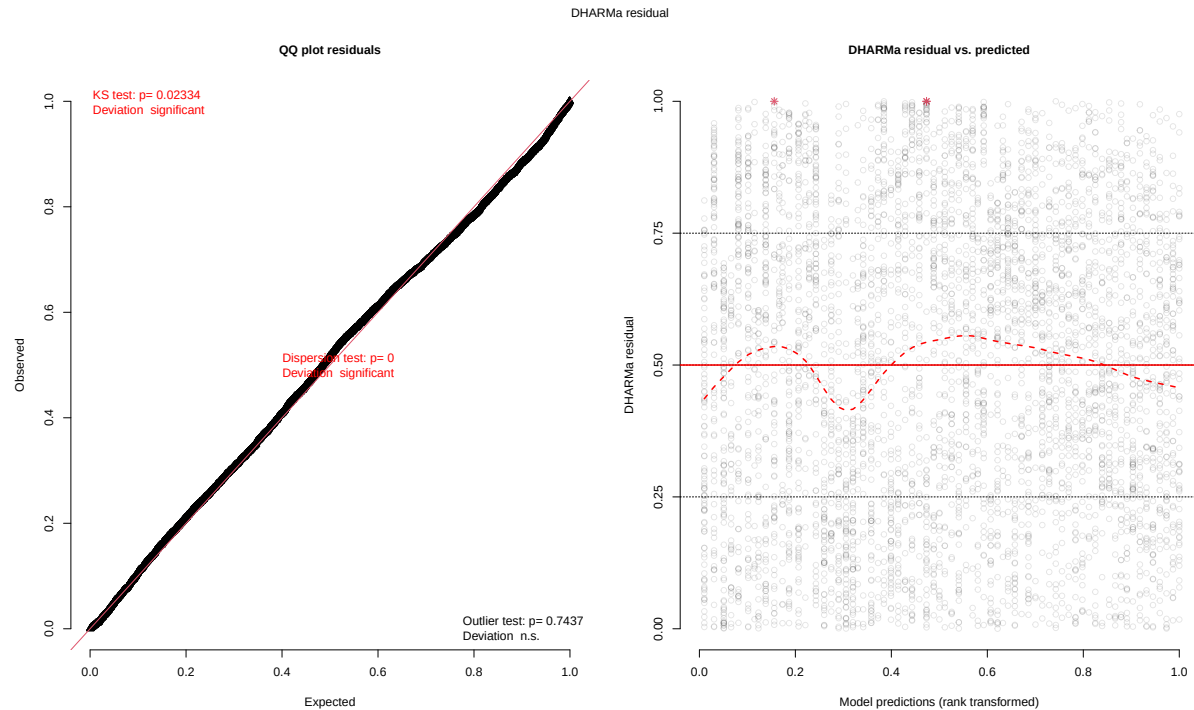
### 3.3 Goodness-of-fit evaluation

Refit the selected model.

```
buds_pois_mod_33 <- glmmTMB(buds ~ week_centered * lineage * rest_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df)
```

Assess goodness-of-fit of the selected model using simulated residuals.

```
buds_pois_mod_33_res <- simulateResiduals(buds_pois_mod_33, plot = TRUE)
```



### 3.4 Selected model summary

```
# Releveling of "lineage" to obtain all pairwise comparisons from the same model.

buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_MT")
summary(glmTMB(buds ~ week_centered * lineage * rest_condition + (1 | replicate_unique),
REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * rest_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  12602.1  12763.8   -6276.1   12552.1     4728
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
```

|                                                                    |            |
|--------------------------------------------------------------------|------------|
| ##                                                                 | Estimate   |
| ## (Intercept)                                                     | 0.457269   |
| ## week_centered                                                   | 0.040455   |
| ## lineageHC_MT                                                    | -0.573663  |
| ## lineageHO_SPC                                                   | -0.256052  |
| ## lineageHO_SPT                                                   | -0.092013  |
| ## lineageHO_VLN                                                   | -1.013640  |
| ## lineageHV_GAL                                                   | -0.061518  |
| ## rest_conditionMechanical disruption                             | -0.063490  |
| ## week_centered:lineageHC_MT                                      | 0.227553   |
| ## week_centered:lineageHO_SPC                                     | 0.101016   |
| ## week_centered:lineageHO_SPT                                     | 0.126876   |
| ## week_centered:lineageHO_VLN                                     | 0.178343   |
| ## week_centered:lineageHV_GAL                                     | 0.140067   |
| ## week_centered:rest_conditionMechanical disruption               | -0.007714  |
| ## lineageHC_MT:rest_conditionMechanical disruption                | -0.209042  |
| ## lineageHO_SPC:rest_conditionMechanical disruption               | 0.140629   |
| ## lineageHO_SPT:rest_conditionMechanical disruption               | 0.030166   |
| ## lineageHO_VLN:rest_conditionMechanical disruption               | -0.119088  |
| ## lineageHV_GAL:rest_conditionMechanical disruption               | -0.061672  |
| ## week_centered:lineageHC_MT:rest_conditionMechanical disruption  | -0.007118  |
| ## week_centered:lineageHO_SPC:rest_conditionMechanical disruption | -0.065524  |
| ## week_centered:lineageHO_SPT:rest_conditionMechanical disruption | 0.042134   |
| ## week_centered:lineageHO_VLN:rest_conditionMechanical disruption | 0.034530   |
| ## week_centered:lineageHV_GAL:rest_conditionMechanical disruption | -0.098026  |
| ##                                                                 | Std. Error |
| ## (Intercept)                                                     | 0.044793   |
| ## week_centered                                                   | 0.023019   |
| ## lineageHC_MT                                                    | 0.086495   |
| ## lineageHO_SPC                                                   | 0.066325   |
| ## lineageHO_SPT                                                   | 0.066574   |
| ## lineageHO_VLN                                                   | 0.074958   |
| ## lineageHV_GAL                                                   | 0.064093   |
| ## rest_conditionMechanical disruption                             | 0.057329   |
| ## week_centered:lineageHC_MT                                      | 0.044824   |
| ## week_centered:lineageHO_SPC                                     | 0.034484   |
| ## week_centered:lineageHO_SPT                                     | 0.034150   |
| ## week_centered:lineageHO_VLN                                     | 0.040251   |
| ## week_centered:lineageHV_GAL                                     | 0.033253   |
| ## week_centered:rest_conditionMechanical disruption               | 0.033359   |
| ## lineageHC_MT:rest_conditionMechanical disruption                | 0.125988   |
| ## lineageHO_SPC:rest_conditionMechanical disruption               | 0.083929   |
| ## lineageHO_SPT:rest_conditionMechanical disruption               | 0.084802   |
| ## lineageHO_VLN:rest_conditionMechanical disruption               | 0.103689   |
| ## lineageHV_GAL:rest_conditionMechanical disruption               | 0.084267   |
| ## week_centered:lineageHC_MT:rest_conditionMechanical disruption  | 0.069369   |
| ## week_centered:lineageHO_SPC:rest_conditionMechanical disruption | 0.048808   |
| ## week_centered:lineageHO_SPT:rest_conditionMechanical disruption | 0.048697   |
| ## week_centered:lineageHO_VLN:rest_conditionMechanical disruption | 0.059922   |
| ## week_centered:lineageHV_GAL:rest_conditionMechanical disruption | 0.049032   |
| ##                                                                 | z value    |
| ## (Intercept)                                                     | 10.209     |
| ## week_centered                                                   | 1.757      |
| ## lineageHC_MT                                                    | -6.632     |

```

## lineageHO_SPC -3.861
## lineageHO_SPT -1.382
## lineageHO_VLN -13.523
## lineageHV_GAL -0.960
## rest_conditionMechanical disruption -1.107
## week_centered:lineageHC_MT 5.077
## week_centered:lineageHO_SPC 2.929
## week_centered:lineageHO_SPT 3.715
## week_centered:lineageHO_VLN 4.431
## week_centered:lineageHV_GAL 4.212
## week_centered:rest_conditionMechanical disruption -0.231
## lineageHC_MT:rest_conditionMechanical disruption -1.659
## lineageHO_SPC:rest_conditionMechanical disruption 1.676
## lineageHO_SPT:rest_conditionMechanical disruption 0.356
## lineageHO_VLN:rest_conditionMechanical disruption -1.149
## lineageHV_GAL:rest_conditionMechanical disruption -0.732
## week_centered:lineageHC_MT:rest_conditionMechanical disruption -0.103
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption -1.343
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 0.865
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 0.576
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption -1.999
## Pr(>|z|)
## (Intercept) < 2e-16 ***
## week_centered 0.078845 .
## lineageHC_MT 3.30e-11 ***
## lineageHO_SPC 0.000113 ***
## lineageHO_SPT 0.166935
## lineageHO_VLN < 2e-16 ***
## lineageHV_GAL 0.337142
## rest_conditionMechanical disruption 0.268097
## week_centered:lineageHC_MT 3.84e-07 ***
## week_centered:lineageHO_SPC 0.003397 **
## week_centered:lineageHO_SPT 0.000203 ***
## week_centered:lineageHO_VLN 9.39e-06 ***
## week_centered:lineageHV_GAL 2.53e-05 ***
## week_centered:rest_conditionMechanical disruption 0.817126
## lineageHC_MT:rest_conditionMechanical disruption 0.097072 .
## lineageHO_SPC:rest_conditionMechanical disruption 0.093821 .
## lineageHO_SPT:rest_conditionMechanical disruption 0.722053
## lineageHO_VLN:rest_conditionMechanical disruption 0.250756
## lineageHV_GAL:rest_conditionMechanical disruption 0.464253
## week_centered:lineageHC_MT:rest_conditionMechanical disruption 0.918272
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption 0.179432
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 0.386910
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 0.564451
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption 0.045582 *
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_SPC")
summary(glmTMB(buds ~ week_centered * lineage * rest_condition + (1 | replicate_unique),
REML = TRUE, family = poisson, data = buds_df))

```

```
## Family: poisson ( log )
```

```

## Formula:
## buds ~ week_centered * lineage * rest_condition + (1 | replicate_unique)
## Data: buds_df
##
##           AIC           BIC      logLik -2*log(L)  df.resid
##    12602.1    12763.8    -6276.1    12552.1      4728
##
## Random effects:
##
## Conditional model:
##   Groups          Name          Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
##                                     Estimate
## (Intercept)                        0.20122
## week_centered                      0.14147
## lineageHO_MT                      0.25605
## lineageHC_MT                      -0.31761
## lineageHO_SPT                     0.16404
## lineageHO_VLN                     -0.75759
## lineageHV_GAL                     0.19453
## rest_conditionMechanical disruption 0.07714
## week_centered:lineageHO_MT        -0.10102
## week_centered:lineageHC_MT         0.12654
## week_centered:lineageHO_SPT        0.02586
## week_centered:lineageHO_VLN        0.07733
## week_centered:lineageHV_GAL        0.03905
## week_centered:rest_conditionMechanical disruption -0.07324
## lineageHO_MT:rest_conditionMechanical disruption -0.14063
## lineageHC_MT:rest_conditionMechanical disruption -0.34967
## lineageHO_SPT:rest_conditionMechanical disruption -0.11046
## lineageHO_VLN:rest_conditionMechanical disruption -0.25972
## lineageHV_GAL:rest_conditionMechanical disruption -0.20230
## week_centered:lineageHO_MT:rest_conditionMechanical disruption 0.06552
## week_centered:lineageHC_MT:rest_conditionMechanical disruption 0.05841
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 0.10766
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 0.10005
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption -0.03250
##
##                                     Std. Error
## (Intercept)                        0.04895
## week_centered                      0.02568
## lineageHO_MT                      0.06633
## lineageHC_MT                      0.08872
## lineageHO_SPT                     0.06944
## lineageHO_VLN                     0.07751
## lineageHV_GAL                     0.06706
## rest_conditionMechanical disruption 0.06130
## week_centered:lineageHO_MT        0.03448
## week_centered:lineageHC_MT        0.04624
## week_centered:lineageHO_SPT       0.03599
## week_centered:lineageHO_VLN       0.04183
## week_centered:lineageHV_GAL       0.03514

```

```

## week_centered:rest_conditionMechanical disruption 0.03563
## lineageHO_MT:rest_conditionMechanical disruption 0.08393
## lineageHC_MT:rest_conditionMechanical disruption 0.12785
## lineageHO_SPT:rest_conditionMechanical disruption 0.08753
## lineageHO_VLN:rest_conditionMechanical disruption 0.10594
## lineageHV_GAL:rest_conditionMechanical disruption 0.08701
## week_centered:lineageHO_MT:rest_conditionMechanical disruption 0.04881
## week_centered:lineageHC_MT:rest_conditionMechanical disruption 0.07049
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 0.05028
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 0.06121
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption 0.05060
## z value
## (Intercept) 4.111
## week_centered 5.510
## lineageHO_MT 3.861
## lineageHC_MT -3.580
## lineageHO_SPT 2.362
## lineageHO_VLN -9.774
## lineageHV_GAL 2.901
## rest_conditionMechanical disruption 1.258
## week_centered:lineageHO_MT -2.929
## week_centered:lineageHC_MT 2.736
## week_centered:lineageHO_SPT 0.718
## week_centered:lineageHO_VLN 1.849
## week_centered:lineageHV_GAL 1.111
## week_centered:rest_conditionMechanical disruption -2.056
## lineageHO_MT:rest_conditionMechanical disruption -1.676
## lineageHC_MT:rest_conditionMechanical disruption -2.735
## lineageHO_SPT:rest_conditionMechanical disruption -1.262
## lineageHO_VLN:rest_conditionMechanical disruption -2.452
## lineageHV_GAL:rest_conditionMechanical disruption -2.325
## week_centered:lineageHO_MT:rest_conditionMechanical disruption 1.343
## week_centered:lineageHC_MT:rest_conditionMechanical disruption 0.829
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 2.141
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 1.634
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption -0.642
## Pr(>|z|)
## (Intercept) 3.94e-05 ***
## week_centered 3.59e-08 ***
## lineageHO_MT 0.000113 ***
## lineageHC_MT 0.000344 ***
## lineageHO_SPT 0.018155 *
## lineageHO_VLN < 2e-16 ***
## lineageHV_GAL 0.003721 **
## rest_conditionMechanical disruption 0.208241
## week_centered:lineageHO_MT 0.003397 **
## week_centered:lineageHC_MT 0.006213 **
## week_centered:lineageHO_SPT 0.472475
## week_centered:lineageHO_VLN 0.064496 .
## week_centered:lineageHV_GAL 0.266492
## week_centered:rest_conditionMechanical disruption 0.039815 *
## lineageHO_MT:rest_conditionMechanical disruption 0.093821 .
## lineageHC_MT:rest_conditionMechanical disruption 0.006238 **
## lineageHO_SPT:rest_conditionMechanical disruption 0.206966

```

```
## lineageHO_VLN:rest_conditionMechanical disruption 0.014220 *
## lineageHV_GAL:rest_conditionMechanical disruption 0.020070 *
## week_centered:lineageHO_MT:rest_conditionMechanical disruption 0.179432
## week_centered:lineageHC_MT:rest_conditionMechanical disruption 0.407332
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 0.032252 *
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 0.102159
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption 0.520692
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_SPT")
summary(glmmTMB(buds ~ week_centered * lineage * rest_condition + (1 | replicate_unique),
REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * rest_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
## 12602.1 12763.8 -6276.1 12552.1      4728
##
## Random effects:
##
## Conditional model:
## Groups          Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
## (Intercept) 0.365256
## week_centered 0.167331
## lineageHO_SPC -0.164039
## lineageHO_MT 0.092013
## lineageHC_MT -0.481650
## lineageHO_VLN -0.921627
## lineageHV_GAL 0.030494
## rest_conditionMechanical disruption -0.033324
## week_centered:lineageHO_SPC -0.025860
## week_centered:lineageHO_MT -0.126876
## week_centered:lineageHC_MT 0.100677
## week_centered:lineageHO_VLN 0.051467
## week_centered:lineageHV_GAL 0.013190
## week_centered:rest_conditionMechanical disruption 0.034420
## lineageHO_SPC:rest_conditionMechanical disruption 0.110464
## lineageHO_MT:rest_conditionMechanical disruption -0.030166
## lineageHC_MT:rest_conditionMechanical disruption -0.239207
## lineageHO_VLN:rest_conditionMechanical disruption -0.149254
## lineageHV_GAL:rest_conditionMechanical disruption -0.091837
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption -0.107659
## week_centered:lineageHO_MT:rest_conditionMechanical disruption -0.042134
## week_centered:lineageHC_MT:rest_conditionMechanical disruption -0.049252
```

```

## week_centered:lineageHO_VLN:rest_conditionMechanical disruption -0.007605
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption -0.140161
##                                                                 Std. Error
## (Intercept)                                                                 0.049287
## week_centered                                                                 0.025225
## lineageHO_SPC                                                                0.069436
## lineageHO_MT                                                                 0.066574
## lineageHC_MT                                                                0.088903
## lineageHO_VLN                                                                0.077724
## lineageHV_GAL                                                                0.067302
## rest_conditionMechanical disruption 0.062486
## week_centered:lineageHO_SPC                                                  0.035994
## week_centered:lineageHO_MT                                                  0.034150
## week_centered:lineageHC_MT                                                  0.045995
## week_centered:lineageHO_VLN                                                 0.041552
## week_centered:lineageHV_GAL                                                 0.034816
## week_centered:rest_conditionMechanical disruption 0.035476
## lineageHO_SPC:rest_conditionMechanical disruption 0.087534
## lineageHO_MT:rest_conditionMechanical disruption 0.084802
## lineageHC_MT:rest_conditionMechanical disruption 0.128442
## lineageHO_VLN:rest_conditionMechanical disruption 0.106626
## lineageHV_GAL:rest_conditionMechanical disruption 0.087829
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption 0.050278
## week_centered:lineageHO_MT:rest_conditionMechanical disruption 0.048697
## week_centered:lineageHC_MT:rest_conditionMechanical disruption 0.070412
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 0.061126
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption 0.050497
##                                                                 z value
## (Intercept)                                                                 7.411
## week_centered                                                                6.633
## lineageHO_SPC                                                                -2.362
## lineageHO_MT                                                                1.382
## lineageHC_MT                                                                -5.418
## lineageHO_VLN                                                                -11.858
## lineageHV_GAL                                                                0.453
## rest_conditionMechanical disruption -0.533
## week_centered:lineageHO_SPC                                                  -0.718
## week_centered:lineageHO_MT                                                  -3.715
## week_centered:lineageHC_MT                                                  2.189
## week_centered:lineageHO_VLN                                                 1.239
## week_centered:lineageHV_GAL                                                 0.379
## week_centered:rest_conditionMechanical disruption 0.970
## lineageHO_SPC:rest_conditionMechanical disruption 1.262
## lineageHO_MT:rest_conditionMechanical disruption -0.356
## lineageHC_MT:rest_conditionMechanical disruption -1.862
## lineageHO_VLN:rest_conditionMechanical disruption -1.400
## lineageHV_GAL:rest_conditionMechanical disruption -1.046
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption -2.141
## week_centered:lineageHO_MT:rest_conditionMechanical disruption -0.865
## week_centered:lineageHC_MT:rest_conditionMechanical disruption -0.699
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption -0.124
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption -2.776
##                                                                 Pr(>|z|)
## (Intercept)                                                                1.26e-13 ***

```

```

## week_centered 3.28e-11 ***
## lineageHO_SPC 0.018155 *
## lineageHO_MT 0.166935
## lineageHC_MT 6.04e-08 ***
## lineageHO_VLN < 2e-16 ***
## lineageHV_GAL 0.650481
## rest_conditionMechanical disruption 0.593822
## week_centered:lineageHO_SPC 0.472475
## week_centered:lineageHO_MT 0.000203 ***
## week_centered:lineageHC_MT 0.028607 *
## week_centered:lineageHO_VLN 0.215487
## week_centered:lineageHV_GAL 0.704790
## week_centered:rest_conditionMechanical disruption 0.331927
## lineageHO_SPC:rest_conditionMechanical disruption 0.206966
## lineageHO_MT:rest_conditionMechanical disruption 0.722053
## lineageHC_MT:rest_conditionMechanical disruption 0.062549 .
## lineageHO_VLN:rest_conditionMechanical disruption 0.161578
## lineageHV_GAL:rest_conditionMechanical disruption 0.295727
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption 0.032252 *
## week_centered:lineageHO_MT:rest_conditionMechanical disruption 0.386910
## week_centered:lineageHC_MT:rest_conditionMechanical disruption 0.484246
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 0.900992
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption 0.005510 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_VLN")
summary(glmmTMB(buds ~ week_centered * lineage * rest_condition + (1 | replicate_unique),
REML = TRUE, family = poisson, data = buds_df))

```

```

## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * rest_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
## 12602.1 12763.8 -6276.1 12552.1    4728
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
## (Intercept) -0.556370
## week_centered 0.218798
## lineageHO_SPT 0.921627
## lineageHO_SPC 0.757588
## lineageHO_MT 1.013640
## lineageHC_MT 0.439977

```

|                                                                    |            |
|--------------------------------------------------------------------|------------|
| ## lineageHV_GAL                                                   | 0.952121   |
| ## rest_conditionMechanical disruption                             | -0.182578  |
| ## week_centered:lineageHO_SPT                                     | -0.051467  |
| ## week_centered:lineageHO_SPC                                     | -0.077327  |
| ## week_centered:lineageHO_MT                                      | -0.178343  |
| ## week_centered:lineageHC_MT                                      | 0.049210   |
| ## week_centered:lineageHV_GAL                                     | -0.038276  |
| ## week_centered:rest_conditionMechanical disruption               | 0.026816   |
| ## lineageHO_SPT:rest_conditionMechanical disruption               | 0.149254   |
| ## lineageHO_SPC:rest_conditionMechanical disruption               | 0.259718   |
| ## lineageHO_MT:rest_conditionMechanical disruption                | 0.119088   |
| ## lineageHC_MT:rest_conditionMechanical disruption                | -0.089953  |
| ## lineageHV_GAL:rest_conditionMechanical disruption               | 0.057417   |
| ## week_centered:lineageHO_SPT:rest_conditionMechanical disruption | 0.007605   |
| ## week_centered:lineageHO_SPC:rest_conditionMechanical disruption | -0.100054  |
| ## week_centered:lineageHO_MT:rest_conditionMechanical disruption  | -0.034530  |
| ## week_centered:lineageHC_MT:rest_conditionMechanical disruption  | -0.041648  |
| ## week_centered:lineageHV_GAL:rest_conditionMechanical disruption | -0.132556  |
| ##                                                                 | Std. Error |
| ## (Intercept)                                                     | 0.060131   |
| ## week_centered                                                   | 0.033019   |
| ## lineageHO_SPT                                                   | 0.077724   |
| ## lineageHO_SPC                                                   | 0.077511   |
| ## lineageHO_MT                                                    | 0.074958   |
| ## lineageHC_MT                                                    | 0.095343   |
| ## lineageHV_GAL                                                   | 0.075608   |
| ## rest_conditionMechanical disruption                             | 0.086399   |
| ## week_centered:lineageHO_SPT                                     | 0.041552   |
| ## week_centered:lineageHO_SPC                                     | 0.041827   |
| ## week_centered:lineageHO_MT                                      | 0.040251   |
| ## week_centered:lineageHC_MT                                      | 0.050688   |
| ## week_centered:lineageHV_GAL                                     | 0.040814   |
| ## week_centered:rest_conditionMechanical disruption               | 0.049778   |
| ## lineageHO_SPT:rest_conditionMechanical disruption               | 0.106626   |
| ## lineageHO_SPC:rest_conditionMechanical disruption               | 0.105936   |
| ## lineageHO_MT:rest_conditionMechanical disruption                | 0.103689   |
| ## lineageHC_MT:rest_conditionMechanical disruption                | 0.141613   |
| ## lineageHV_GAL:rest_conditionMechanical disruption               | 0.106192   |
| ## week_centered:lineageHO_SPT:rest_conditionMechanical disruption | 0.061126   |
| ## week_centered:lineageHO_SPC:rest_conditionMechanical disruption | 0.061215   |
| ## week_centered:lineageHO_MT:rest_conditionMechanical disruption  | 0.059922   |
| ## week_centered:lineageHC_MT:rest_conditionMechanical disruption  | 0.078592   |
| ## week_centered:lineageHV_GAL:rest_conditionMechanical disruption | 0.061383   |
| ##                                                                 | z value    |
| ## (Intercept)                                                     | -9.253     |
| ## week_centered                                                   | 6.626      |
| ## lineageHO_SPT                                                   | 11.858     |
| ## lineageHO_SPC                                                   | 9.774      |
| ## lineageHO_MT                                                    | 13.523     |
| ## lineageHC_MT                                                    | 4.615      |
| ## lineageHV_GAL                                                   | 12.593     |
| ## rest_conditionMechanical disruption                             | -2.113     |
| ## week_centered:lineageHO_SPT                                     | -1.239     |
| ## week_centered:lineageHO_SPC                                     | -1.849     |

```

## week_centered:lineageHO_MT -4.431
## week_centered:lineageHC_MT 0.971
## week_centered:lineageHV_GAL -0.938
## week_centered:rest_conditionMechanical disruption 0.539
## lineageHO_SPT:rest_conditionMechanical disruption 1.400
## lineageHO_SPC:rest_conditionMechanical disruption 2.452
## lineageHO_MT:rest_conditionMechanical disruption 1.149
## lineageHC_MT:rest_conditionMechanical disruption -0.635
## lineageHV_GAL:rest_conditionMechanical disruption 0.541
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 0.124
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption -1.634
## week_centered:lineageHO_MT:rest_conditionMechanical disruption -0.576
## week_centered:lineageHC_MT:rest_conditionMechanical disruption -0.530
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption -2.159
## Pr(>|z|)
## (Intercept) < 2e-16 ***
## week_centered 3.44e-11 ***
## lineageHO_SPT < 2e-16 ***
## lineageHO_SPC < 2e-16 ***
## lineageHO_MT < 2e-16 ***
## lineageHC_MT 3.94e-06 ***
## lineageHV_GAL < 2e-16 ***
## rest_conditionMechanical disruption 0.0346 *
## week_centered:lineageHO_SPT 0.2155
## week_centered:lineageHO_SPC 0.0645 .
## week_centered:lineageHO_MT 9.39e-06 ***
## week_centered:lineageHC_MT 0.3316
## week_centered:lineageHV_GAL 0.3483
## week_centered:rest_conditionMechanical disruption 0.5901
## lineageHO_SPT:rest_conditionMechanical disruption 0.1616
## lineageHO_SPC:rest_conditionMechanical disruption 0.0142 *
## lineageHO_MT:rest_conditionMechanical disruption 0.2508
## lineageHC_MT:rest_conditionMechanical disruption 0.5253
## lineageHV_GAL:rest_conditionMechanical disruption 0.5887
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 0.9010
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption 0.1022
## week_centered:lineageHO_MT:rest_conditionMechanical disruption 0.5645
## week_centered:lineageHC_MT:rest_conditionMechanical disruption 0.5962
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption 0.0308 *
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

buds_df$lineage <- relevel(buds_df$lineage, ref = "HC_MT")
summary(glmTMB(buds ~ week_centered * lineage * rest_condition + (1 | replicate_unique),
REML = TRUE, family = poisson, data = buds_df))

```

```

## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * rest_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  12602.1  12763.8   -6276.1   12552.1     4728

```

```

##
## Random effects:
##
## Conditional model:
##   Groups          Name          Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##                                     Estimate
## (Intercept)                       -0.116393
## week_centered                      0.268008
## lineageHO_VLN                     -0.439977
## lineageHO_SPT                      0.481650
## lineageHO_SPC                      0.317611
## lineageHO_MT                      0.573663
## lineageHV_GAL                     0.512144
## rest_conditionMechanical disruption -0.272531
## week_centered:lineageHO_VLN        -0.049210
## week_centered:lineageHO_SPT        -0.100677
## week_centered:lineageHO_SPC        -0.126537
## week_centered:lineageHO_MT        -0.227553
## week_centered:lineageHV_GAL        -0.087487
## week_centered:rest_conditionMechanical disruption -0.014832
## lineageHO_VLN:rest_conditionMechanical disruption 0.089953
## lineageHO_SPT:rest_conditionMechanical disruption 0.239207
## lineageHO_SPC:rest_conditionMechanical disruption 0.349671
## lineageHO_MT:rest_conditionMechanical disruption 0.209042
## lineageHV_GAL:rest_conditionMechanical disruption 0.147370
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 0.041648
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 0.049252
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption -0.058406
## week_centered:lineageHO_MT:rest_conditionMechanical disruption 0.007118
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption -0.090908
##                                     Std. Error
## (Intercept)                       0.074015
## week_centered                      0.038461
## lineageHO_VLN                     0.095343
## lineageHO_SPT                     0.088903
## lineageHO_SPC                     0.088717
## lineageHO_MT                     0.086495
## lineageHV_GAL                     0.087060
## rest_conditionMechanical disruption 0.112202
## week_centered:lineageHO_VLN        0.050688
## week_centered:lineageHO_SPT        0.045995
## week_centered:lineageHO_SPC        0.046244
## week_centered:lineageHO_MT        0.044824
## week_centered:lineageHV_GAL        0.045329
## week_centered:rest_conditionMechanical disruption 0.060821
## lineageHO_VLN:rest_conditionMechanical disruption 0.141613
## lineageHO_SPT:rest_conditionMechanical disruption 0.128442
## lineageHO_SPC:rest_conditionMechanical disruption 0.127850
## lineageHO_MT:rest_conditionMechanical disruption 0.125988
## lineageHV_GAL:rest_conditionMechanical disruption 0.128148

```

```

## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 0.078592
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 0.070412
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption 0.070488
## week_centered:lineageHO_MT:rest_conditionMechanical disruption 0.069369
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption 0.070639
## z value
## (Intercept) -1.573
## week_centered 6.968
## lineageHO_VLN -4.615
## lineageHO_SPT 5.418
## lineageHO_SPC 3.580
## lineageHO_MT 6.632
## lineageHV_GAL 5.883
## rest_conditionMechanical disruption -2.429
## week_centered:lineageHO_VLN -0.971
## week_centered:lineageHO_SPT -2.189
## week_centered:lineageHO_SPC -2.736
## week_centered:lineageHO_MT -5.077
## week_centered:lineageHV_GAL -1.930
## week_centered:rest_conditionMechanical disruption -0.244
## lineageHO_VLN:rest_conditionMechanical disruption 0.635
## lineageHO_SPT:rest_conditionMechanical disruption 1.862
## lineageHO_SPC:rest_conditionMechanical disruption 2.735
## lineageHO_MT:rest_conditionMechanical disruption 1.659
## lineageHV_GAL:rest_conditionMechanical disruption 1.150
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 0.530
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 0.699
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption -0.829
## week_centered:lineageHO_MT:rest_conditionMechanical disruption 0.103
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption -1.287
## Pr(>|z|)
## (Intercept) 0.115820
## week_centered 3.21e-12 ***
## lineageHO_VLN 3.94e-06 ***
## lineageHO_SPT 6.04e-08 ***
## lineageHO_SPC 0.000344 ***
## lineageHO_MT 3.30e-11 ***
## lineageHV_GAL 4.04e-09 ***
## rest_conditionMechanical disruption 0.015143 *
## week_centered:lineageHO_VLN 0.331626
## week_centered:lineageHO_SPT 0.028607 *
## week_centered:lineageHO_SPC 0.006213 **
## week_centered:lineageHO_MT 3.84e-07 ***
## week_centered:lineageHV_GAL 0.053602 .
## week_centered:rest_conditionMechanical disruption 0.807336
## lineageHO_VLN:rest_conditionMechanical disruption 0.525295
## lineageHO_SPT:rest_conditionMechanical disruption 0.062549 .
## lineageHO_SPC:rest_conditionMechanical disruption 0.006238 **
## lineageHO_MT:rest_conditionMechanical disruption 0.097072 .
## lineageHV_GAL:rest_conditionMechanical disruption 0.250145
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 0.596166
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 0.484246
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption 0.407332
## week_centered:lineageHO_MT:rest_conditionMechanical disruption 0.918272

```

```
## week_centered:lineageHV_GAL:rest_conditionMechanical disruption 0.198115
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HV_GAL")
summary(glmmTMB(buds ~ week_centered * lineage * rest_condition + (1 | replicate_unique),
REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * rest_condition + (1 | replicate_unique)
## Data: buds_df
##
##          AIC          BIC    logLik -2*log(L)  df.resid
##   12602.1   12763.8   -6276.1   12552.1     4728
##
## Random effects:
##
## Conditional model:
## Groups          Name          Variance Std.Dev.
## replicate_unique (Intercept) 0.03111  0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
##                                     Estimate
## (Intercept)                        0.39575
## week_centered                      0.18052
## lineageHC_MT                      -0.51214
## lineageHO_VLN                     -0.95212
## lineageHO_SPT                     -0.03049
## lineageHO_SPC                     -0.19453
## lineageHO_MT                      0.06152
## rest_conditionMechanical disruption -0.12516
## week_centered:lineageHC_MT         0.08749
## week_centered:lineageHO_VLN        0.03828
## week_centered:lineageHO_SPT       -0.01319
## week_centered:lineageHO_SPC       -0.03905
## week_centered:lineageHO_MT       -0.14007
## week_centered:rest_conditionMechanical disruption -0.10574
## lineageHC_MT:rest_conditionMechanical disruption -0.14737
## lineageHO_VLN:rest_conditionMechanical disruption -0.05742
## lineageHO_SPT:rest_conditionMechanical disruption 0.09184
## lineageHO_SPC:rest_conditionMechanical disruption 0.20230
## lineageHO_MT:rest_conditionMechanical disruption 0.06167
## week_centered:lineageHC_MT:rest_conditionMechanical disruption 0.09091
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 0.13256
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 0.14016
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption 0.03250
## week_centered:lineageHO_MT:rest_conditionMechanical disruption 0.09803
##
##                                     Std. Error
## (Intercept)                        0.04592
## week_centered                      0.02400
## lineageHC_MT                      0.08706
```

|                                                                    |              |
|--------------------------------------------------------------------|--------------|
| ## lineageHO_VLN                                                   | 0.07561      |
| ## lineageHO_SPT                                                   | 0.06730      |
| ## lineageHO_SPC                                                   | 0.06706      |
| ## lineageHO_MT                                                    | 0.06409      |
| ## rest_conditionMechanical disruption                             | 0.06174      |
| ## week_centered:lineageHC_MT                                      | 0.04533      |
| ## week_centered:lineageHO_VLN                                     | 0.04081      |
| ## week_centered:lineageHO_SPT                                     | 0.03482      |
| ## week_centered:lineageHO_SPC                                     | 0.03514      |
| ## week_centered:lineageHO_MT                                      | 0.03325      |
| ## week_centered:rest_conditionMechanical disruption               | 0.03593      |
| ## lineageHC_MT:rest_conditionMechanical disruption                | 0.12815      |
| ## lineageHO_VLN:rest_conditionMechanical disruption               | 0.10619      |
| ## lineageHO_SPT:rest_conditionMechanical disruption               | 0.08783      |
| ## lineageHO_SPC:rest_conditionMechanical disruption               | 0.08701      |
| ## lineageHO_MT:rest_conditionMechanical disruption                | 0.08427      |
| ## week_centered:lineageHC_MT:rest_conditionMechanical disruption  | 0.07064      |
| ## week_centered:lineageHO_VLN:rest_conditionMechanical disruption | 0.06138      |
| ## week_centered:lineageHO_SPT:rest_conditionMechanical disruption | 0.05050      |
| ## week_centered:lineageHO_SPC:rest_conditionMechanical disruption | 0.05060      |
| ## week_centered:lineageHO_MT:rest_conditionMechanical disruption  | 0.04903      |
| ##                                                                 | z value      |
| ## (Intercept)                                                     | 8.618        |
| ## week_centered                                                   | 7.522        |
| ## lineageHC_MT                                                    | -5.883       |
| ## lineageHO_VLN                                                   | -12.593      |
| ## lineageHO_SPT                                                   | -0.453       |
| ## lineageHO_SPC                                                   | -2.901       |
| ## lineageHO_MT                                                    | 0.960        |
| ## rest_conditionMechanical disruption                             | -2.027       |
| ## week_centered:lineageHC_MT                                      | 1.930        |
| ## week_centered:lineageHO_VLN                                     | 0.938        |
| ## week_centered:lineageHO_SPT                                     | -0.379       |
| ## week_centered:lineageHO_SPC                                     | -1.111       |
| ## week_centered:lineageHO_MT                                      | -4.212       |
| ## week_centered:rest_conditionMechanical disruption               | -2.943       |
| ## lineageHC_MT:rest_conditionMechanical disruption                | -1.150       |
| ## lineageHO_VLN:rest_conditionMechanical disruption               | -0.541       |
| ## lineageHO_SPT:rest_conditionMechanical disruption               | 1.046        |
| ## lineageHO_SPC:rest_conditionMechanical disruption               | 2.325        |
| ## lineageHO_MT:rest_conditionMechanical disruption                | 0.732        |
| ## week_centered:lineageHC_MT:rest_conditionMechanical disruption  | 1.287        |
| ## week_centered:lineageHO_VLN:rest_conditionMechanical disruption | 2.159        |
| ## week_centered:lineageHO_SPT:rest_conditionMechanical disruption | 2.776        |
| ## week_centered:lineageHO_SPC:rest_conditionMechanical disruption | 0.642        |
| ## week_centered:lineageHO_MT:rest_conditionMechanical disruption  | 1.999        |
| ##                                                                 | Pr(> z )     |
| ## (Intercept)                                                     | < 2e-16 ***  |
| ## week_centered                                                   | 5.38e-14 *** |
| ## lineageHC_MT                                                    | 4.04e-09 *** |
| ## lineageHO_VLN                                                   | < 2e-16 ***  |
| ## lineageHO_SPT                                                   | 0.65048      |
| ## lineageHO_SPC                                                   | 0.00372 **   |
| ## lineageHO_MT                                                    | 0.33714      |

```
## rest_conditionMechanical disruption 0.04265 *
## week_centered:lineageHC_MT 0.05360 .
## week_centered:lineageHO_VLN 0.34833
## week_centered:lineageHO_SPT 0.70479
## week_centered:lineageHO_SPC 0.26649
## week_centered:lineageHO_MT 2.53e-05 ***
## week_centered:rest_conditionMechanical disruption 0.00325 **
## lineageHC_MT:rest_conditionMechanical disruption 0.25015
## lineageHO_VLN:rest_conditionMechanical disruption 0.58872
## lineageHO_SPT:rest_conditionMechanical disruption 0.29573
## lineageHO_SPC:rest_conditionMechanical disruption 0.02007 *
## lineageHO_MT:rest_conditionMechanical disruption 0.46425
## week_centered:lineageHC_MT:rest_conditionMechanical disruption 0.19812
## week_centered:lineageHO_VLN:rest_conditionMechanical disruption 0.03081 *
## week_centered:lineageHO_SPT:rest_conditionMechanical disruption 0.00551 **
## week_centered:lineageHO_SPC:rest_conditionMechanical disruption 0.52069
## week_centered:lineageHO_MT:rest_conditionMechanical disruption 0.04558 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## 4 Estimated marginal means

### 4.1 Table

Create a character string containing the 6 exact centered values.

```
week_values <- 1:6 - mean(buds_df$week, na.rm = TRUE)
week_terms <- paste0("week_centered [", paste(week_values, collapse = ","), "]")
```

Compute the estimated marginal means of weekly bud production for every combination of week, rest condition, and lineage, based on the selected model.

```
buds_emmeans <-
  as.data.frame(ggemmeans(buds_pois_mod_33,
    terms = c(week_terms, "rest_condition", "lineage"),
    bias_correction = TRUE)) %>%
  arrange(facet, group, x)
```

Back-transform the centered week to the original scale (weeks 1 to 6) for plotting.

```
buds_emmeans$x <- buds_emmeans$x + mean(buds_df$week, na.rm = TRUE)
```

Round the values to ensure ggplot perfectly recognizes the integers 1 to 6.

```
buds_emmeans$x <- round(buds_emmeans$x)
```

### 4.2 Plots

### 4.3 Common functions

Function to plot, for a single lineage, the predicted weekly bud production by rest condition.

```

buds_plot_treatment <- function(lineage, show_y_text = FALSE, show_x_text = TRUE,
show_y_title = FALSE, show_x_title = FALSE) {
  data_sub <- subset(buds_emmeans, facet == lineage)

  prefix <- sub("_.*", "", lineage)
  prefix_format <- paste0(substr(prefix, 1, 1), tolower(substr(prefix, 2,
nchar(prefix))))
  label_expr <- parse(text = paste0("bolditalic('", prefix_format, "') * bold('",
sub(".*_", "_", lineage), "')"))

  ggplot(data_sub, aes(x = x, y = predicted, color = group, fill = group)) +
    geom_line(linewidth = 0.8) +
    geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, color = NA) +
    labs(x = if(show_x_title) "Age (weeks)" else NULL,
         y = if(show_y_title) "Estimated marginal means\nof the number of buds produced"
         else NULL,
         title = NULL) +
    theme_bw() +
    theme(axis.text.x = if(show_x_text) element_text(color = "black", size = 14) else
element_blank(),
          axis.ticks.x = if(show_x_text) element_line() else element_blank(),
          axis.text.y = if(show_y_text) element_text(color = "black", size = 14) else
element_blank(),
          axis.ticks.y = if(show_y_text) element_line() else element_blank(),
          axis.title.x = element_text(color = "black", size = 14, margin = margin(t =
10)),
          axis.title.y.right = element_text(color = "black", size = 14, angle = 90,
margin = margin(l = 25)),
          axis.line.y.right = element_blank(),
          plot.margin = margin(l = 5, r = 5, t = 5, b = 5),
          legend.position = "none") +
    scale_x_continuous(limits = c(1, 6), breaks = c(2, 4, 6)) +
    scale_y_continuous(position = "right") +
    coord_cartesian(ylim = c(0, 3.5)) +
    scale_color_manual(values = rest_condition_colors) +
    scale_fill_manual(values = rest_condition_colors) +
    annotate("text", x = 1.2, y = 3.3, label = label_expr,
            color = "black", size = 5, hjust = 0, vjust = 1) }

```

Function to plot, for a single lineage, the predicted weekly bud production pooled across rest conditions.

```

buds_plot_lineage <- function(lineage, show_y_text = FALSE, show_x_text = TRUE,
show_y_title = FALSE, show_x_title = FALSE) {
  data_sub <- subset(buds_emmeans, facet == lineage)
  data_lineage <- aggregate(cbind(predicted, conf.low, conf.high) ~ x, data = data_sub,
FUN = mean)
  data_lineage <- data_lineage[order(data_lineage$x), ]

  prefix <- sub("_.*", "", lineage)
  prefix_format <- paste0(substr(prefix, 1, 1), tolower(substr(prefix, 2,
nchar(prefix))))
  label_expr <- parse(text = paste0("bolditalic('", prefix_format, "') * bold('",
sub(".*_", "_", lineage), "')"))

```

```

ggplot(data_lineage, aes(x = x, y = predicted)) +
  geom_line(linewidth = 0.8, color = "black") +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, fill = "black",
    linewidth = 0.3) +
  labs(x = if(show_x_title) "Age (weeks)" else NULL,
    y = if(show_y_title) "Estimated marginal means\nof the number of buds produced"
    else NULL,
    title = NULL) +
  theme_bw() +
  theme(axis.text.x = if(show_x_text) element_text(color = "black", size = 14) else
    element_blank(),
    axis.ticks.x = if(show_x_text) element_line() else element_blank(),
    axis.text.y = if(show_y_text) element_text(color = "black", size = 14) else
    element_blank(),
    axis.ticks.y = if(show_y_text) element_line() else element_blank(),
    axis.title.x = element_text(color = "black", size = 14, margin = margin(t =
    10)),
    axis.title.y.right = element_text(color = "black", size = 14, angle = 90,
    margin = margin(l = 25)),
    axis.line.y.right = element_blank(),
    plot.margin = margin(l = 5, r = 5, t = 5, b = 5)) +
  scale_x_continuous(limits = c(1, 6), breaks = c(2, 4, 6)) +
  scale_y_continuous(position = "right") +
  coord_cartesian(ylim = c(0, 3.5)) +
  annotate("text", x = 1.2, y = 3.3, label = label_expr,
    color = "black", size = 5, hjust = 0, vjust = 1) }

```

#### 4.4 Panel A: interaction between rest condition and lineage

```

A1 <- buds_plot_treatment("HO_MT", show_y_text = FALSE, show_x_text = FALSE)
A2 <- buds_plot_treatment("HO_SPC", show_y_text = FALSE, show_x_text = FALSE)
A3 <- buds_plot_treatment("HO_SPT", show_y_text = TRUE, show_x_text = FALSE)
A4 <- buds_plot_treatment("HO_VLN", show_y_text = FALSE, show_x_text = TRUE)
A5 <- buds_plot_treatment("HC_MT", show_y_text = FALSE, show_x_text = TRUE, show_x_title
= TRUE)
A6 <- buds_plot_treatment("HV_GAL", show_y_text = TRUE, show_x_text = TRUE, show_y_title
= TRUE)

```

Add a letter tag (A) directly to the first sub-plot before combining.

```

A1 <- A1 + labs(tag = "A") + theme(plot.tag = element_text(size = 20, face = "bold"))

```

Combine A1 to A6 into panel A.

```

A <- (A1 + A2 + A3) / (A4 + A5 + A6) + plot_layout(heights = c(1, 1))

```

#### 4.5 Panel B: lineage effect

```

B1 <- buds_plot_lineage("HO_MT", show_y_text = FALSE, show_x_text = FALSE)
B2 <- buds_plot_lineage("HO_SPC", show_y_text = FALSE, show_x_text = FALSE)
B3 <- buds_plot_lineage("HO_SPT", show_y_text = TRUE, show_x_text = FALSE)
B4 <- buds_plot_lineage("HO_VLN", show_y_text = FALSE, show_x_text = TRUE)
B5 <- buds_plot_lineage("HC_MT", show_y_text = FALSE, show_x_text = TRUE, show_x_title = TRUE)
B6 <- buds_plot_lineage("HV_GAL", show_y_text = TRUE, show_x_text = TRUE, show_y_title = TRUE)

```

Add a letter tag (B) directly to the first sub-plot before combining.

```

B1 <- B1 + labs(tag = "B") + theme(plot.tag = element_text(size = 20, face = "bold"))

```

Combine B1 to B6 into panel B.

```

B <- (B1 + B2 + B3) / (B4 + B5 + B6) + plot_layout(heights = c(1, 1))

```

## 4.6 Assembly

Extract legend from the plot A6 to use later on the full figure.

```

legend_treatment <- cowplot::get_legend(
  A6 +
  labs(color = NULL, fill = NULL) +
  theme(legend.position = "bottom",
        legend.direction = "horizontal",
        legend.text = element_text(size = 13)))

```

Build title graphical objects for each panel.

```

interaction_title_grob <- cowplot::ggdraw() +
  cowplot::draw_label("Interaction between mechanical disruption and lineage",
                      fontface = "bold", size = 16, x = 0.5, hjust = 0.5)

lineage_title_grob <- cowplot::ggdraw() +
  cowplot::draw_label("Lineage effect",
                      fontface = "bold", size = 16, x = 0.5, hjust = 0.5)

```

Stack the titles, panels, and legend vertically into a single column.

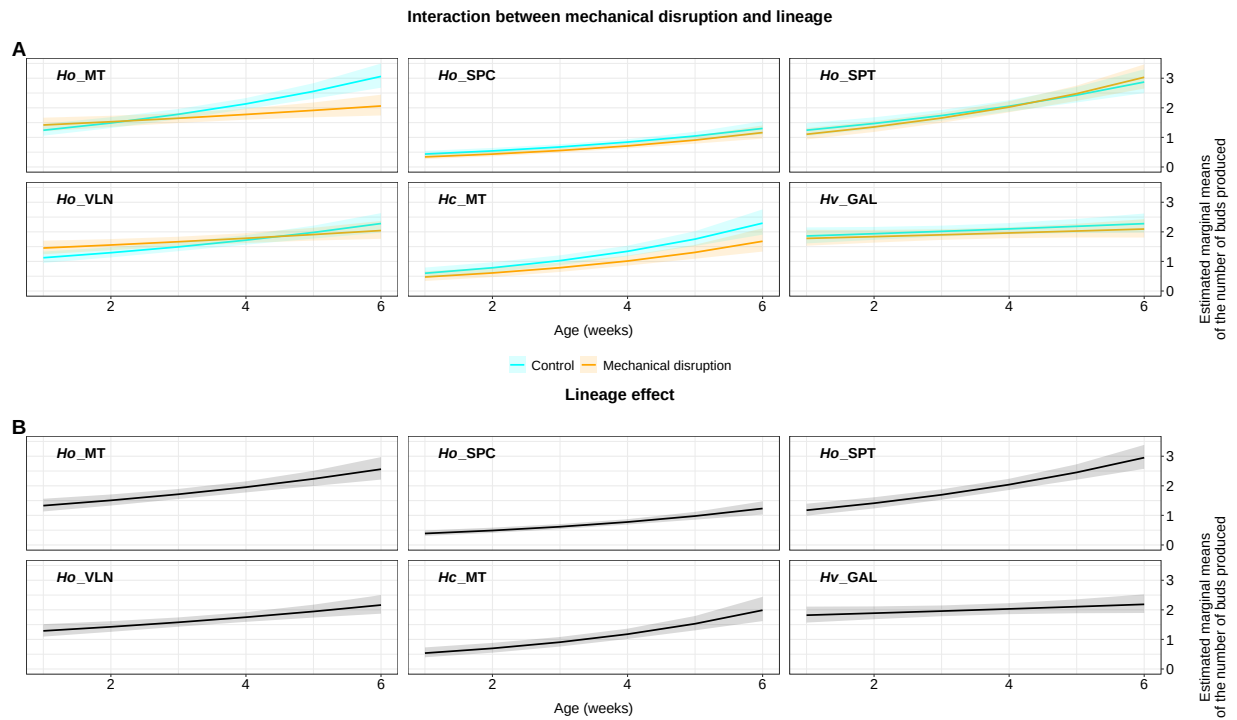
```

final_buds_panel <- cowplot::plot_grid(interaction_title_grob,
                                       A,
                                       legend_treatment,
                                       lineage_title_grob,
                                       B,
                                       ncol = 1,
                                       rel_heights = c(0.08, 1, 0.1, 0.08, 1)) +
  theme(plot.margin = margin(l = 5, r = 5, t = 5, b = 5))

```

Show the final figure.

```
print(final_buds_panel)
```



```
ggsave("figure_5.png", plot = final_buds_panel, width = 21, height = 29.7, units = "cm",  
dpi = 300)
```